

$$a). |\psi_1\rangle \otimes |\psi_2\rangle = \frac{1}{2} (|0R\rangle|0R\rangle + |0R\rangle|1B\rangle + |1B\rangle|0R\rangle + |1B\rangle|1B\rangle)$$

$$= \frac{1}{2} (|0R0R\rangle + |0R1B\rangle + |1B0R\rangle + |1B1B\rangle)$$

$$|00\rangle = \frac{1}{\sqrt{2}} (|\Phi_A^+\rangle + |\Phi_A^-\rangle)$$

$$|11\rangle = \frac{1}{\sqrt{2}} (|\Phi_A^+\rangle - |\Phi_A^-\rangle)$$

$$|01\rangle = \frac{1}{\sqrt{2}} (|\Psi_A^+\rangle + |\Psi_A^-\rangle)$$

$$|10\rangle = \frac{1}{\sqrt{2}} (|\Psi_A^+\rangle - |\Psi_A^-\rangle)$$

$$|RR\rangle = \frac{1}{\sqrt{2}} (|\Phi_{EM}^+\rangle + |\Phi_{EM}^-\rangle)$$

$$|BB\rangle = \frac{1}{\sqrt{2}} (|\Phi_{EM}^+\rangle - |\Phi_{EM}^-\rangle)$$

$$|RB\rangle = \frac{1}{\sqrt{2}} (|\Psi_{EM}^+\rangle + |\Psi_{EM}^-\rangle)$$

$$|BR\rangle = \frac{1}{\sqrt{2}} (|\Psi_{EM}^+\rangle - |\Psi_{EM}^-\rangle)$$

$$|\psi_1\rangle \otimes |\psi_2\rangle = \frac{1}{2} (|\Phi_A^+\rangle \otimes |\Phi_{EM}^+\rangle + |\Phi_A^+\rangle \otimes |\Phi_{EM}^-\rangle + |\Phi_A^-\rangle \otimes |\Psi_{EM}^+\rangle + |\Phi_A^-\rangle \otimes |\Psi_{EM}^-\rangle)$$

b).

$$P = \frac{1}{2} (|RB\rangle - |BR\rangle) (\langle RB| - \langle BR|)$$

$$P|\psi\rangle = \frac{1}{2} (|\Phi_A^-\rangle \otimes |\Psi_{EM}^-\rangle)$$

$$|\psi_{Atom}\rangle = \frac{1}{\sqrt{2}} (|01\rangle - |10\rangle)$$

$$P_A = \text{Tr}_B (|\psi_A\rangle \langle \psi_A|) = \frac{1}{2} I$$

$$S(P_A) = -\text{Tr}(P_A \log P_A) = -(\frac{1}{2}(-1) + \frac{1}{2}(-1)) = 1$$

c). Let's say the unitary for the 50/50 Beam Splitter is:

$$\hat{a}_b \rightarrow \frac{1}{\sqrt{2}} (\hat{a}_{D1}^\dagger + \hat{a}_{D2}^\dagger)$$

$$\hat{a}_B^\dagger \rightarrow \frac{1}{\sqrt{2}} (\hat{a}_{D1}^\dagger - \hat{a}_{D2}^\dagger)$$

$$|RB\rangle \rightarrow \hat{a}_R^\dagger \hat{a}_B^\dagger \rightarrow \frac{1}{2} (|2,0\rangle - |0,2\rangle + 2|1,1\rangle)$$

$$|BR\rangle \rightarrow \frac{1}{2} (|2,0\rangle - |0,2\rangle)$$

$$|BB\rangle \rightarrow \frac{1}{2} (|2,0\rangle - 2|1,1\rangle + |0,2\rangle)$$

$$|RR\rangle \rightarrow \frac{1}{2} (|2,0\rangle + 2|1,1\rangle + |0,2\rangle)$$

$$\text{for } |\psi_{EM}^+\rangle: |RB\rangle + |BR\rangle = |2,0\rangle - |0,2\rangle + 2|1,1\rangle$$

$$\text{for } |\psi_{EM}^-\rangle: |1,1\rangle$$

$$\text{for } |\Phi^\pm\rangle: |2,0\rangle / |0,2\rangle$$

therefore only $|\psi_{EM}^-\rangle$ ensure $|1,1\rangle$ on the detector